



User Manual

MMU Energy Prediction Simulator (EPS)

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1 Introduction

This MMU Energy Prediction Software (EPS) is used to predict MMU energy consumption (kwh). This can be used as system fault detection and diagnosis to analyze the energy consumption and increase the efficiency of energy usage.

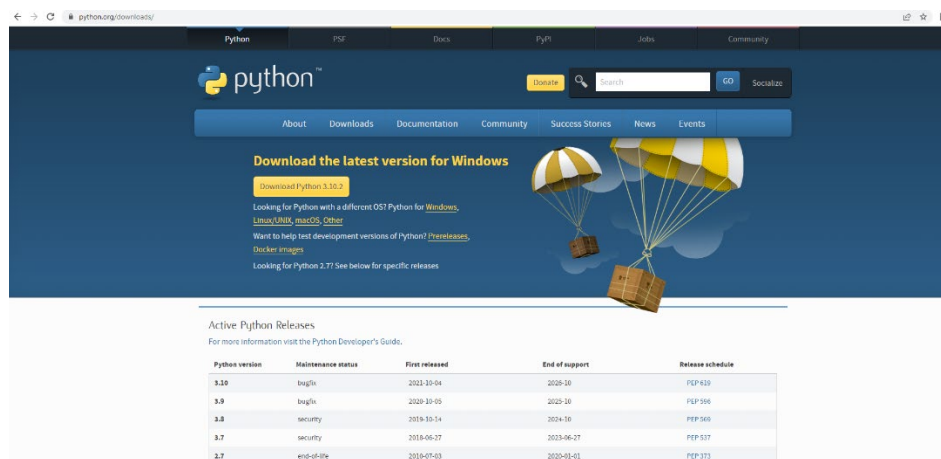
1.1 Key Features of MMU Energy Prediction Software (EPS)

- Simple, minimalistic and user-friendly interface
- Easy to use on different computers
- Fairly accurate prediction based on MMU historical data of energy consumption

2 Getting Started

1.1 Downloading the necessary modules and library to run the software

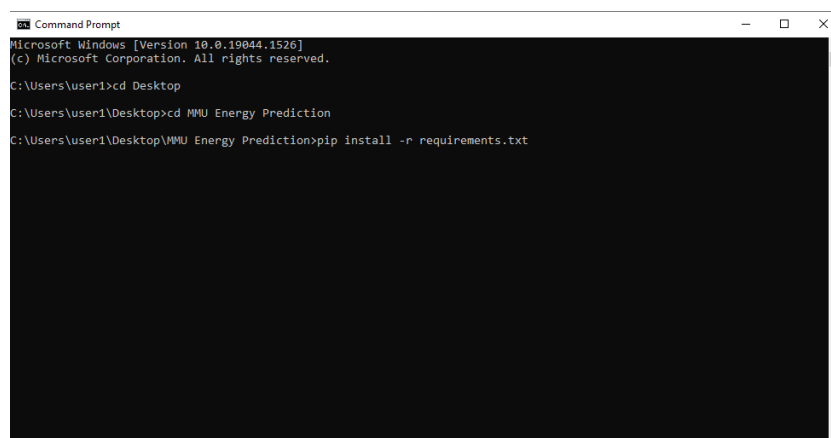
1. Install python programming language. Navigate to the Python website, <https://www.python.org/>.



2. There will be a requirements.txt file inside the folder provided. Open the command prompt and run the following line:

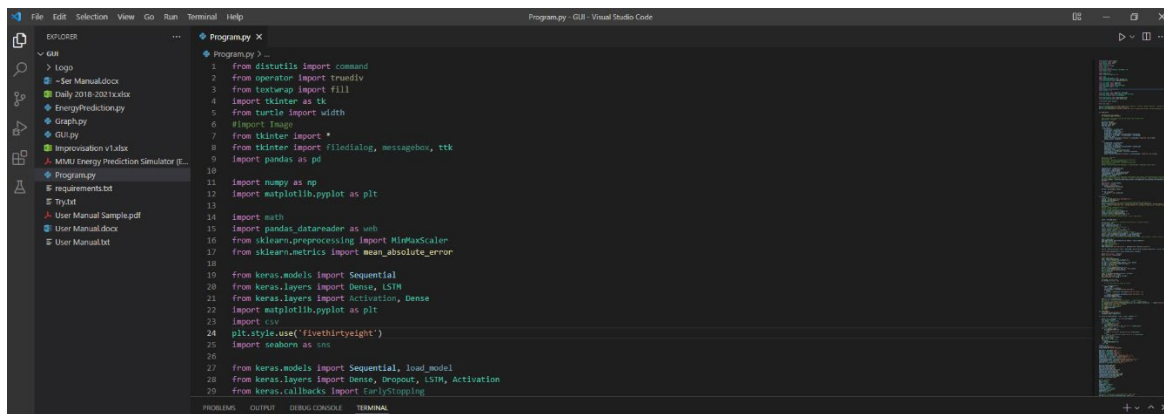
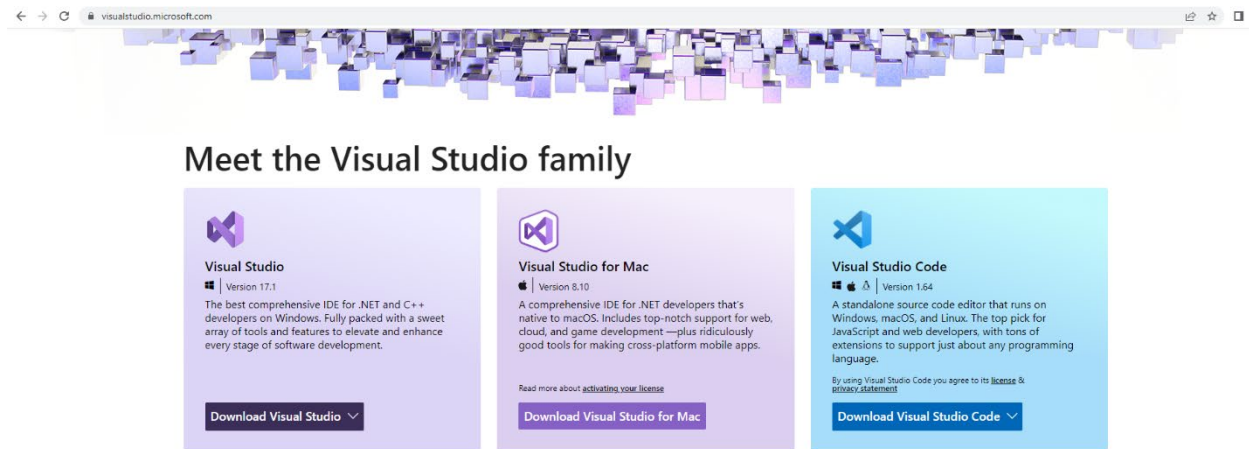
Do note that the path of the file should be directed first before running the line.

```
pip install -r requirements.txt
```



Optional

- This is an optional software to see the code behind the software. Navigate to the Visual Studio website, <https://visualstudio.microsoft.com/>. Download the Visual Studio Code.



[illegible]

4. Below are the table of guidelines to enter the proper value. The max prediction date is **August 31, 2022**.

Variables	Value
Day	1 - 31
Month	1 - 12
Year	2019 - 2022
Forecasted temperature (°C)	22 - 31
Forecasted humidity (%)	58 – 95
Forecasted pressure (hPa)	1006 - 1014
Forecasted rainfall duration (min)	0 - 60
Forecasted rainfall amount (mm)	-14 - 4
Forecasted wind speed (m/s)	0 – 5.3
Type of day	0 for weekdays and 1 for weekend and holiday
Type of lockdown	0 for no MCO, 1 for MCO and 2 for RMCO

5. Below are the examples of how the variables value are correctly input. Click the **Enter** button to store the current values and the **Clear** button to clear all the variables that has been stored.

MMU Energy Prediction

Day (DD)* Month (MM)* Year (YYYY)*

Date : 7 8 2022

Type of lockdown * Temperature (°C) * Relative Humidity (%) * Pressure (hPa) *

2 28 80 1011

Rainfall Duration (min) * Rainfall amount (mm) * Wind speed (m/s) * Type of day *

0 0 1 1

Enter Clear

Database is successfully updated

TimeStamp	Usage Peak (kwh)	Average pressure (Hg)	Average temperature
1/8/2022	4831.356196943973	1011.0	28.0
2/8/2022	4831.356196943973	1011.0	28.0
3/8/2022	4831.356196943973	1011.0	28.0
4/8/2022	4831.356196943973	1011.0	28.0
5/8/2022	4831.356196943973	1011.0	28.0
6/8/2022	3751.269354838709	1011.0	28.0
7/8/2022	3751.269354838709	1011.0	28.0

Train Update

Predicted Data

- Press the **Train** button to train the prediction model with historical data allocated in the folder and show the prediction of the values that has been input.

```

Command Prompt - python Program.py

[2 rows x 9 columns]
3/5/2022
2021-03-05 00:00:00
  TimeStamp  Usage  Peak (kwh)  Average pressure (Hg)  ...  Rainfall duration (min)  Rainfall amount (mm)  Type of day
0  1/5/2022    3902.322581    1011  ...             45             5             0
1  2/5/2022    3902.322581    1011  ...             45             5             0
2  3/5/2022    5088.322581    1011  ...             45             5             0

[3 rows x 9 columns]
4/5/2022
2021-04-05 00:00:00
  TimeStamp  Usage  Peak (kwh)  Average pressure (Hg)  ...  Rainfall duration (min)  Rainfall amount (mm)  Type of day
0  1/5/2022    3902.322581    1011  ...             45             5             0
1  2/5/2022    3902.322581    1011  ...             45             5             0
2  3/5/2022    5088.322581    1011  ...             45             5             0
3  4/5/2022    5088.322581    1011  ...             45             5             0

[4 rows x 9 columns]
5/5/2022
2021-05-05 00:00:00
  TimeStamp  Usage  Peak (kwh)  Average pressure (Hg)  ...  Rainfall duration (min)  Rainfall amount (mm)  Type of day
0  1/5/2022    3902.322581    1011  ...             45             5             0
1  2/5/2022    3902.322581    1011  ...             45             5             0
2  3/5/2022    5088.322581    1011  ...             45             5             0
3  4/5/2022    5088.322581    1011  ...             45             5             0
4  5/5/2022    5088.322581    1011  ...             45             5             0

[5 rows x 9 columns]
  
```

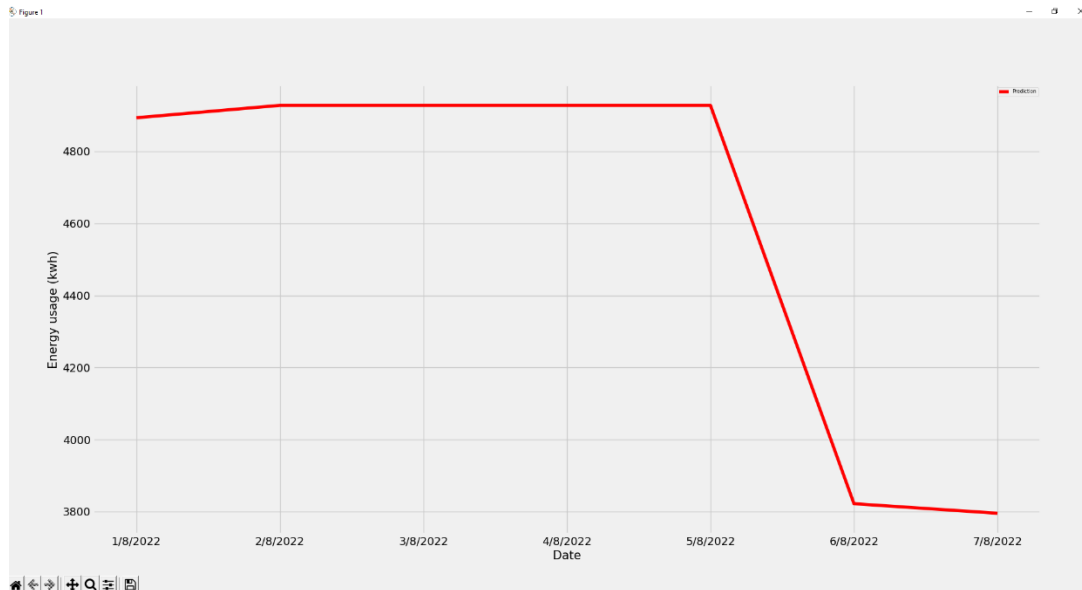
- You can monitor the progress of the model training in the command prompt.

```

Command Prompt - python Program.py

: 0.8000
Epoch 1868/2000
19/19 [=====] - 0s 4ms/step - loss: 0.0245 - accuracy: 0.9931 - val_loss: 0.2495 - val_accuracy
: 0.8000
Epoch 1869/2000
19/19 [=====] - 0s 4ms/step - loss: 0.0233 - accuracy: 0.9931 - val_loss: 0.2784 - val_accuracy
: 0.8000
Epoch 1870/2000
19/19 [=====] - 0s 4ms/step - loss: 0.0239 - accuracy: 0.9931 - val_loss: 0.2704 - val_accuracy
: 0.8000
Epoch 1871/2000
19/19 [=====] - 0s 4ms/step - loss: 0.0244 - accuracy: 0.9923 - val_loss: 0.2681 - val_accuracy
: 0.8000
Epoch 1872/2000
19/19 [=====] - 0s 4ms/step - loss: 0.0238 - accuracy: 0.9931 - val_loss: 0.2666 - val_accuracy
: 0.8000
Epoch 1873/2000
19/19 [=====] - 0s 4ms/step - loss: 0.0234 - accuracy: 0.9931 - val_loss: 0.2618 - val_accuracy
: 0.8000
Epoch 1874/2000
19/19 [=====] - 0s 5ms/step - loss: 0.0235 - accuracy: 0.9931 - val_loss: 0.2630 - val_accuracy
: 0.8000
Epoch 1875/2000
19/19 [=====] - 0s 4ms/step - loss: 0.0235 - accuracy: 0.9923 - val_loss: 0.2599 - val_accuracy
: 0.8000
Epoch 1876/2000
19/19 [=====] - 0s 4ms/step - loss: 0.0223 - accuracy: 0.9931 - val_loss: 0.2607 - val_accuracy
: 0.8000
Epoch 1877/2000
19/19 [=====] - 0s 4ms/step - loss: 0.0228 - accuracy: 0.9931 - val_loss: 0.2487 - val_accuracy
  
```

8. After the model is done trained for approximately **5 minutes**, a graph showing the prediction of energy consumption (kwh) on the inputted forecasted date is plotted.



9. Press the Update button to show the predicted values on the table titled, “Predicted Data”.

MMU Energy Prediction

Day (DD)*: 7 Month (MM)*: 8 Year (YYYY)*: 2022

Type of lockdown *: 2 Temperature (°C) *: 28 Relative Humidity (%) *: 80 Pressure (hPa) *: 1011

Rainfall Duration (min) *: 0 Rainfall amount (mm) *: 0 Wind speed (m/s) *: 1 Type of day *: 1

Enter Clear

Database is successfully updated

TimeStamp	Usage Peak (kwh)	Average pressure (Hg)	Average temperature
1/8/2022	4831.356196943973	1011.0	28.0
2/8/2022	4831.356196943973	1011.0	28.0
3/8/2022	4831.356196943973	1011.0	28.0
4/8/2022	4831.356196943973	1011.0	28.0
5/8/2022	4831.356196943973	1011.0	28.0
6/8/2022	3751.269354838709	1011.0	28.0
7/8/2022	3751.269354838709	1011.0	28.0

Train Update

TimeStamp	Predicted Usage Peak (kwh)
1/8/2022	4893.32763671875
2/8/2022	4927.740234375
3/8/2022	4927.740234375
4/8/2022	4927.740234375
5/8/2022	4927.740234375
6/8/2022	3821.97021484375
7/8/2022	3795.483642578125